Lumen: A Tangible Somaesthetic Peripheral for Negotiated Screen-Work Interruption

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Fig. 1. Lumen in three states from left to right: *calm* (deep white pulsing), *break* (deep red pulsing), and *rest* (deep green pulsing after user tap) states.

Interruption coordination is a longstanding HCI concern, yet many deployed systems rely on immediate interruption: system-timed prompts that demand focal attention regardless of the user's attentional or bodily state. A somaesthetic alternative is needed that signals attentional state change while preserving the user's agency to respond. In this demo, we present *Lumen*, a tangible peripheral object designed as an instance of negotiated interruption. Lumen is a hand-cast translucent silicone dome that sits on the desk and pulses in response to the body's blink rhythm via a webcam using the Eye Aspect Ratio method. Three ambient breathing states are rendered through variations in light rhythm and color, reflecting different blink-derived attentional needs. Designed through a soma-informed iterative process, the interaction aims to remain subtle, peripheral, and bodily perceivable rather than an explicit notification. An exploratory showcase with five participants suggested that the interaction was perceived as bodily evocative and identified the legibility of the negotiation gesture as a emerging design concern for future ambient interaction research.

Additional Key Words and Phrases: Negotiated interruption, Tangible interaction, Peripheral ambient display, Ambient light, Blink rate sensing, Somaesthetic design

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1 Introduction

Sustained screen work suppresses the involuntary blinking rhythm of the body: a healthy rate of 15–20 blinks per minute drops to 4-5 during focused work [8], which contributes to the symptoms of Computer Vision Syndrome, affecting up to 69% of screen users [1]. Prevailing digital wellbeing tools, such as Pomodoro timers, notification chimes, and screen-freeze applications, respond to this somatic condition by overriding it. McFarlane's taxonomy [7] distinguishes *immediate interruption* from *negotiated interruption*, in which the system signals state while preserving the user's

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agency to respond. Inspired by the negotiated interruption, we designed a peripheral desktop artifact for negotiating screen-work breaks through ambient light rhythms and simple touch interaction. Using the first-person perspective and soma design methods, we treated blink-derived bodily rhythms as a dynamic design material for shaping the temporal behaviors of an interactive physical artifact, to explore how subtle temporal light behaviors may support non-prescriptive attentional awareness. We further conducted an exploratory showcase with five participants to gather initial reflections on the artifact's embodied qualities.

2 Lumen

Lumen is a hand-cast translucent silicone dome, that sits on the user's desk and glows. It uses webcam-based blink detection (MediaPipe Face Mesh and the Eye Aspect Ratio (EAR) method [9]) to estimate the user's normal blink frequency, and sense prolonged blink suppression, without storing video and connecting to the network. These changes are mapped onto four ambient breathing light behaviors corresponding to different needs for attentional disengagement: stable blinking, reduced but sustainable blinking, prolonged blink suppression, and during rest. We iteratively designed and experienced four corresponding Lumen breathing behaviors as: a slow white breathing light that remains unnoticeable, a slightly faster but still peripheral white breathing light, a red breathing rhythm intended to evoke attentional awareness and mild concern, and a slow green breathing light associated with rest and interaction. An embedded force sensor supports simple, passive, and embodied interactions activated by sustained holding and gentle tapping, allowing users to negotiate transitions between concentrated work and rest.

In the design of Lumen, Soma Design [4] was used as a primary approach to explore how blink-derived bodily rhythms could be expressed through dynamic light behaviors and physical form. Through first-person design inquiry, iterative decisions regarding materiality, interaction, and ambient light behavior were guided by felt bodily experience. The final artifact further draws on concepts from graduated soft interruption [3], ambient-display notification levels [6], tangible interaction [5], and slow technology [2].

3 Reflections and Future Works

Small-scale exploratory evaluations were conducted, including a four-week autoethnographic study, iterative design of light behaviors, informal feedback with colleagues, and a 90-minute showcase with five participants. During the iterative design process, a range of light behaviors was experienced and reflected upon, including asymmetric breathing, stochastic peak brightness, variations in breathing rhythm, brief color shifts during breathing, laggy brightness effects, and subtle jitter. Our somatic feedback guided the iterative design and refinement processes toward perceptually salient yet non-intrusive patterns. Through these exploratory studies, we found that: (1) participants preferred slow, symmetric, and non-colored breathing pulses in screen-focused contexts; (2) complex temporal dynamics were less noticeable than color changes, and colored lights (e.g., red and green) were strongly associated with symbolic meanings such as traffic signals; (3) one participant described the system as "a living organism," and outside screen-focused work contexts, participants expressed interest in richer embodied interactions (e.g., stroking, lifting and replacing, or two-hands embracing). These findings are exploratory and suggest directions for future structured lab and in-situ studies on attention and embodied light aesthetics.

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5 Generative AI Disclosure

The authors used large language models (Claude by Anthropic and ChatGPT by OpenAI) and Grammarly Pro to paraphrase and improve the clarity of selected sentences. All research ideas, design decisions, fabrication, code, and conclusions are solely the authors' own.

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